

ROHIT YELUKATI MAHENDRA

Applied AI Engineer | AI Automation, Intelligent Systems & Digital Transformation

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PROFESSIONAL PROFILE

Applied AI Engineer and sole developer with 5+ years at Xeal Pharma, delivering end-to-end AI, automation, OCR, and operational platforms for regulated pharmaceutical workflows. Combines Python/Django engineering with applied LLM systems, agentic workflows, document intelligence, analytics, and process transformation. Experienced in translating director-level operational needs into secure, production-ready tools that improve traceability, reduce manual work, and support scalable decision-making.

CORE EXPERTISE

Python, Django, FastAPI, LLMs & AI agents, multi-agent workflows, OCR and document intelligence, RAG and structured extraction, Celery/Redis, SQL and analytics, APIs and integrations, Playwright, workflow automation, secure local/hybrid AI, process mapping, digital transformation, regulated pharmaceutical operations

PROFESSIONAL EXPERIENCE

Python / AI Developer | Xeal Pharma

Apr 2023 - Present | Birmingham, UK (Hybrid)

- Sole developer designing and delivering internal AI, automation, OCR, and operational systems from discovery and architecture through release and iteration.
- Built secure local and hybrid AI solutions for sensitive pharmaceutical use cases, including multi-agent workflows, structured clinical summarisation, intelligent extraction, and risk-focused decision support.
- Developed Python/Django backend platforms integrating task queues, document processing, search, analytics, APIs, and operator-facing workflows across clinical support, order intelligence, dispatch, pricing, and reporting.
- Partnered directly with director-level stakeholders to convert operational pain points into production-ready systems that improve traceability, reduce rework, and enable scale.

Python Developer | Xeal Pharma

Aug 2021 - Jun 2023

- Developed automation and operational software foundations for a pharmaceutical environment, spanning data workflows, reporting, process support, and internal tools.
- Progressed into full ownership of AI-enabled internal platforms and end-to-end solution delivery.

Casual Researcher | University of Hertfordshire

Apr 2022 - Jul 2022 | Part-time

- Built a cancer image annotation tool combining CNN-based tumour detection with automated binary mask generation to support medical imaging research workflows.
- Implemented a manual review and correction mode for tumour-level annotation, with structured JSON pipelines storing coordinate metadata for downstream analysis and model development.

Technology Internship Experience | Bright Network

Jul 2020 - Aug 2020

SELECTED AI & PLATFORM PROJECTS

AI CBPM Clinician | Xeal Pharma

Feb 2026 - Present

- Built a fully local, guardrailed multi-agent clinical support system that transforms patient and treatment information into structured summaries and intelligent risk flags while keeping sensitive data on-premises.

DAX Order Intelligence & Automation Platform | Xeal Pharma

Dec 2021 - Present

- Architected a modular, distributed Django platform using Celery, Redis, NGINX, and Gunicorn for scalable order intelligence and process automation.
- Implemented QR-triggered workflow capture, real-time pricing, auditable order traces, productivity reporting, dashboards, rule-based automation, and LLM-powered email extraction and SQL analytics.
- Applied predictive modelling for workforce efficiency and volume forecasting, alongside NLP-assisted document correction and external regulatory data crawling.

OUROBOROS Kernel Mint | Open-source / Hugging Face Build Small Hackathon

2026

- Created a verifier-led system in which a small model generates Triton GPU kernels and automated compilation, correctness checks, and benchmarking determine verified training candidates.

EXPANDED PORTFOLIO & OPEN-SOURCE WORK

- SentinelEval (2025): Full-stack GenAI evaluation and OWASP LLM Top 10 security-testing platform with FastAPI, React, Celery/Redis workers, adversarial scans, telemetry, dashboards, a Python SDK, and a trace proxy.
- Gemma 3 270M fine-tuning (2025): Fine-tuned a local compact model for pharmaceutical product-to-stock-code mapping, achieving 99% accuracy and reducing manual inventory effort.
- WebsiteWhen (2025): Full-stack AI product combining local-business discovery, AI research summaries, prototype landing-page generation, authentication, and map-based exploration.
- KairoScan (2025): LLM-ready Python web-assessment toolkit covering adaptive port scanning, crawling, TLS and headers, injection/XSS testing, API and JavaScript discovery, authentication testing, reporting, and dynamic SPA analysis through Playwright.
- Pharmaceutical Dispatch Automation and PharmaOCR: Automated courier/dispatch workflows; built a Windows-native, YOLOv8 and OCR prescription-processing engine with layout-aware extraction, patient-indexed storage, and offline retrieval.
- OpenSceneSense and OpenSceneSense Ollama: Published Python packages for cloud and local-first multimodal video analysis, covering frame analysis, audio transcription, dynamic frame selection, structured outputs, and summarisation.
- AutoGPT contribution: Shipped upstream Replicate Flux image-generation blocks, including configurable production workflow controls, into the AutoGPT agent framework.
- Additional delivery: UKPostCodeIO client library; payslip email drafting; DAX Reprint and Price Calculator; bespoke supplement-label generation; EEG emotion recognition; cancer-image annotation; Pet Me/See Me smart pet products; QR treasure hunt, grocery comparison, and IoT smart-car apps.

RESEARCH, PUBLICATION & DEVELOPER TOOLING

- Maintains 23 public GitHub repositories and reusable Python packages, with 244 contributions in the past year. Pinned work includes OpenSceneSense, OpenSceneSense Ollama, UKPostCodeIO, and contributions to PyTorch, PyTorch Tutorials, and ExecuTorch.
- MSc research: developed real-time EEG emotion recognition using the DREAMER dataset and Random Forest classification for valence, arousal, and dominance; achieved 92%, 97.5%, and 90% accuracy respectively.

XEAL 2.0 DIGITAL TRANSFORMATION

- Directly supporting the transition from fragmented, paper-based processes to a digitally controlled operating model with traceable records, integrated workflows, and stronger operational visibility.
- Current portfolio includes ERP and eQMS implementation support, Jira and Slack rollout, Confluence documentation revamp, process mapping, security review, cloud-versus-local architecture decisions, order-processing automation, operational reporting, API integrations, dashboards, KPIs/KRIs, risk tracking, and data remediation.

EDUCATION

Master's Degree, Artificial Intelligence and Robotics | University of Hertfordshire

Jan 2020 - Mar 2022

BTech, Computer Science | Vidya Jyothi Institute of Technology

2015 - 2019

CERTIFICATIONS & RECOGNITION

- First Place Contributor, PyTorch Docathon 2026 - 19 merged pull requests across PyTorch, ExecuTorch, Tutorials, and the PyTorch website; 47 points.
- Build Small Hackathon 2026, Hugging Face (issued Jun 2026).
- MTA: Database Fundamentals - Certified 2018, Microsoft (issued Jul 2018).
- Young Innovator, T-Hub (2017) - one of three recipients nationally for the Pet Me! project.

OPEN-SOURCE & PROFESSIONAL CONTRIBUTIONS

- Publishes Python packages and contributes to open source across video understanding, local AI analysis, developer tooling, and AI documentation; earned GitHub Pull Shark, Starstruck, Pair Extraordinaire, Quickdraw, and YOLO achievements.
- Received professional recommendations recognising technical dedication, feature delivery, and sustained contribution to Xeal Pharma.

COMMUNITY & EARLIER PRODUCT WORK

- Hyderabad Zonal Executive, Infotsav (2019): managed registrations and participation across Hyderabad for national hackathon qualifiers.
- Built consumer and IoT products from 2016 onwards, including a smart car, Find My Grocery, Treasure Scan, Pet Me, See Me, and the Pet Me companion app—developing early experience in mobile products, connected devices, computer vision, and user-centred innovation.
- Pet Me was recognised by T-Hub as a Young Innovator project and helped establish a record of taking ideas from concept through prototype and public presentation.